

IPv6 Addressing

INTRODUCTION

An IPv6 address is 128 bits or 16 bytes (octet) long.

The address length in IPv6 is four times of the length address in IPv4.



Notations

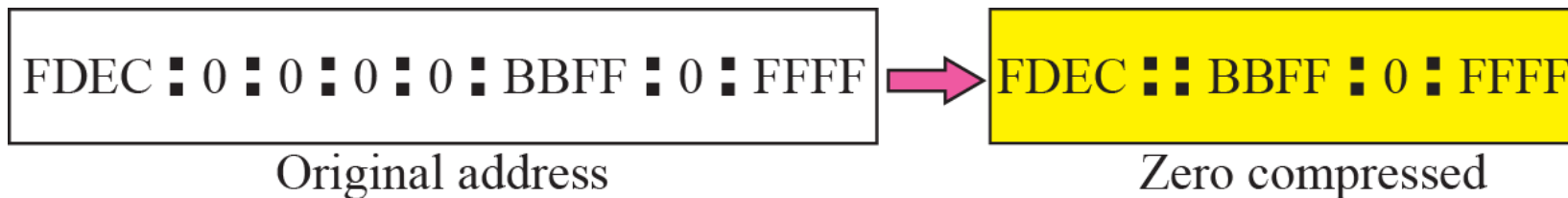
Dotted decimal notation (not used)

221.14.65.11.105.45.170.34.12.234.18.0.14.0.115.255

Colon hexadecimal notation

FDEC : BA98 : 7654 : 3210 : ADBF : BBFF : 2922 : FFFF

*Zero compression
(allowed only once per address)*



CIDR address

FDEC :: BBFF : 0 : FFFF/60

Abbreviations (for simplify representation of leading zeros)

Ex. 0074 – 74 , 000F – F

0000- 0

3210 - 3210

Show the unabbreviated colon hex notation for the following IPv6 addresses:

- a.** An address with 64 0s followed by 64 1s.
- b.** An address with 128 0s.
- c.** An address with 128 1s.
- d.** An address with 128 alternative 1s and 0s.

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Solution

- a.** 0000:0000:0000:0000:FFFF:FFFF:FFFF:FFFF
- b.** 0000:0000:0000:0000:0000:0000:0000:0000
- c.** FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF
- d.** AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA

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the zero contraction version of addresses (part c and d cannot be abbreviated)

- a.** :: FFFF:FFFF:FFFF:FFFF
- b.** ::
- c.** FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF
- d.** AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA:AAAA

Example 26.3

Show abbreviations for the following addresses:

- a.** 0000:0000:FFFF:0000:0000:0000:0000:0000
- b.** 1234:2346:0000:0000:0000:0000:0000:1111
- c.** 0000:0001:0000:0000:0000:0000:1200:1000
- d.** 0000:0000:0000:0000:0000:FFFF:24.123.12.6

Example 26.3

Show abbreviations for the following addresses:

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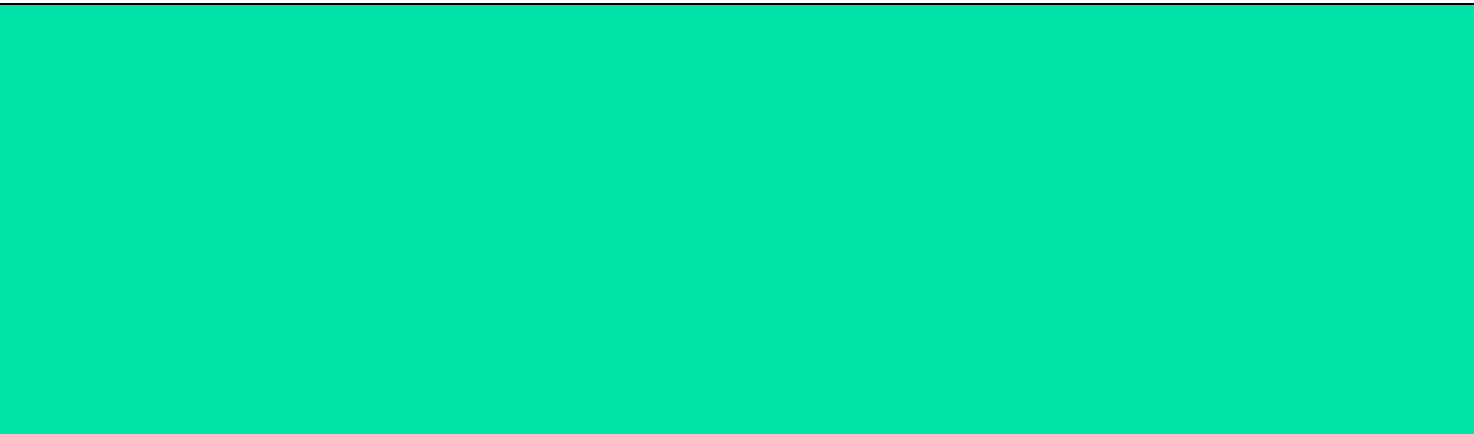
Solution

- a.** 0:0:FFFF::
- b.** 1234:2346::1111
- c.** 0:1::1200:1000
- d.** ::FFFF:24.123.12.6

Example 26.4

Decompress the following addresses and show the complete unabbreviated IPv6 address:

- a.** 1111::2222
- b.** ::
- c.** 0:1::
- d.** AAAA:A:AA::1234



Example 26.4

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- a. 1111::2222
- b. ::
- c. 0:1::
- d. AAAA:A:AA::1234

Solution

- a. 1111:0000:0000:0000:0000:0000:0000:2222
- b. 0000:0000:0000:0000:0000:0000:0000:0000
- c. 0000:0001:0000:0000:0000:0000:0000:0000
- d. AAAA:000A:00AA:0000:0000:0000:0000:1234

Address Space

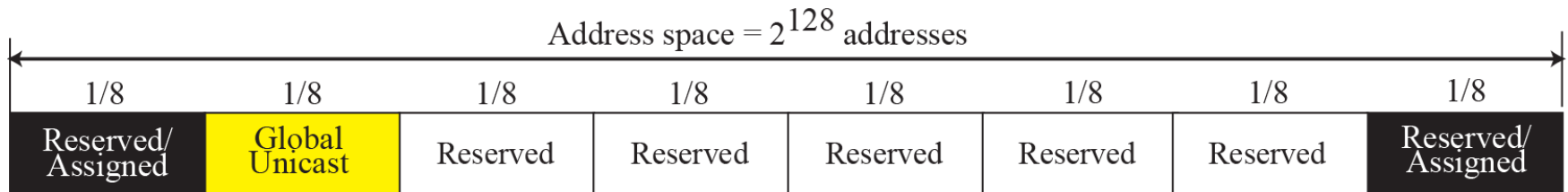
- If we assume that the number of people on the planet earth is soon to be 2^{34} (more than 16 billion). Each person can have 2^{94} addresses to use.
- If we assign 2^{60} addresses to the users each year (almost one billion each second), it takes 2^{68} years to deplete addresses.
- If we can build a high-rise building over the land and sea to accommodate 2^{68} computers in each square meter of the earth, still there are enough addresses to connect all computers to the Internet (the planet earth is approximately 2^{60} square meters).

ADDRESS SPACE ALLOCATION

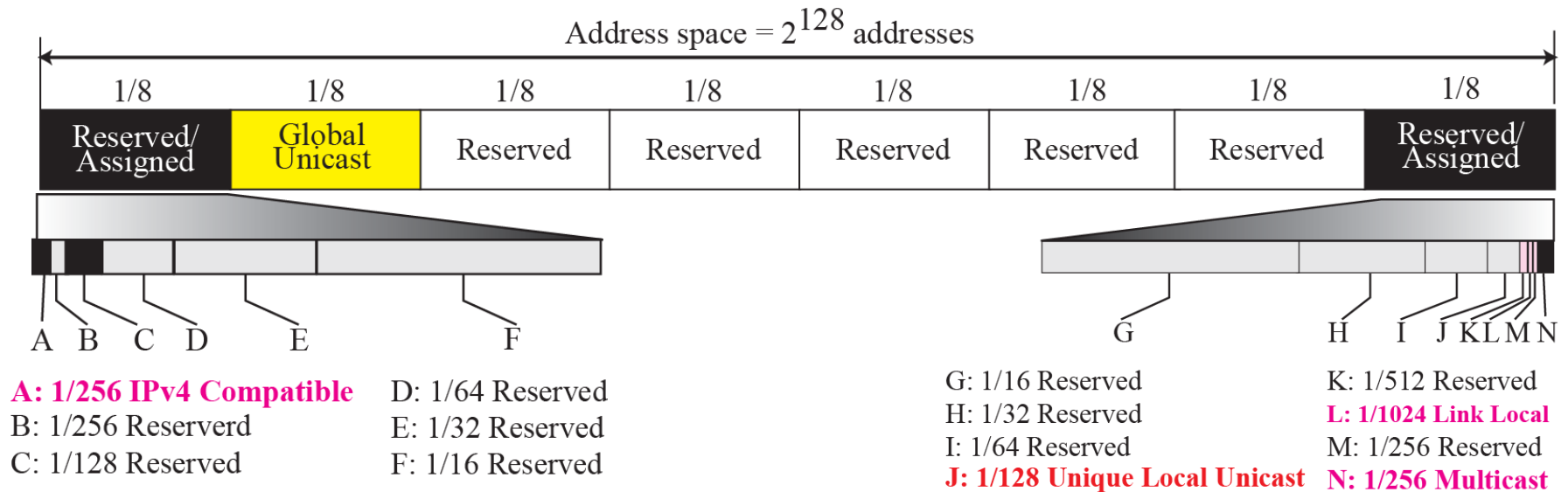
the address space of IPv6 is divided into several blocks of varying size and each block is allocated for special purpose.

Most of the blocks are still unassigned and have been left aside for future use.

- ✓ we first divide the whole address space into eight equal ranges.



Address space allocation



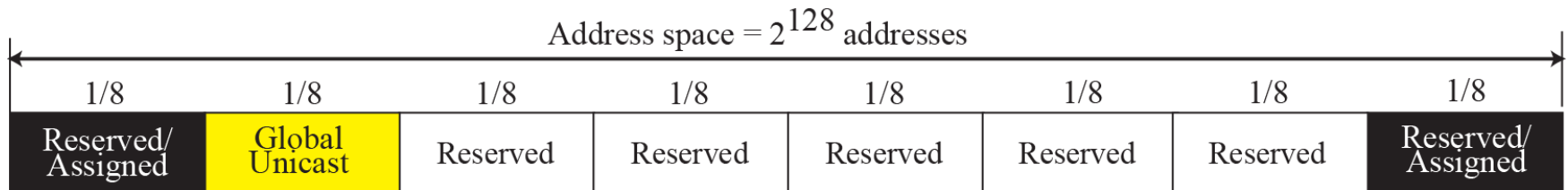
This division does not show the block allocation, but it shows where each actual block is located

Table 26.1 *Prefixes for IPv6 Addresses*

	<i>Block Prefix</i>	<i>CIDR</i>	<i>Block Assignment</i>	<i>Fraction</i>
1	0000 0000	0000::/8	Reserved (IPv4 compatible)	1/256
	0000 0001	0100::/8	Reserved	1/256
	0000 001	0200::/7	Reserved	1/128
	0000 01	0400::/6	Reserved	1/64
	0000 1	0800::/5	Reserved	1/32
	0001	1000::/4	Reserved	1/16
2	001	2000::/3	Global unicast	1/8
3	010	4000::/3	Reserved	1/8
4	011	6000::/3	Reserved	1/8
5	100	8000::/3	Reserved	1/8
6	101	A000::/3	Reserved	1/8
7	110	C000::/3	Reserved	1/8
8	1110	E000::/4	Reserved	1/16
	1111 0	F000::/5	Reserved	1/32
	1111 10	F800::/6	Reserved	1/64
	1111 110	FC00::/7	Unique local unicast	1/128
	1111 1110 0	FE00::/9	Reserved	1/512
	1111 1110 10	FE80::/10	Link local addresses	1/1024
	1111 1110 11	FEC0::/10	Reserved	1/1024
	1111 1111	FF00::/8	Multicast addresses	1/256

Example

Address Space Allocation shows that only a portion of the address space can be used for global unicast communication. How many addresses are in this block?



Example

Address Space Allocation shows that only a portion of the address space can be used for global unicast communication. How many addresses are in this block?

Solution

This block occupies only one-eighth of the address spaces. To find the number of addresses, we can divide the total address space by 8 or 2^3 .

The result is $(2^{128})/(2^3) = 2^{125}$ —a huge block.

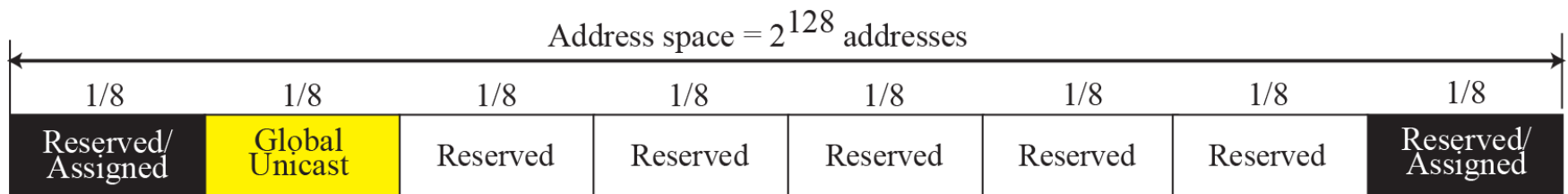


Table 26.1 Prefixes for IPv6 Addresses

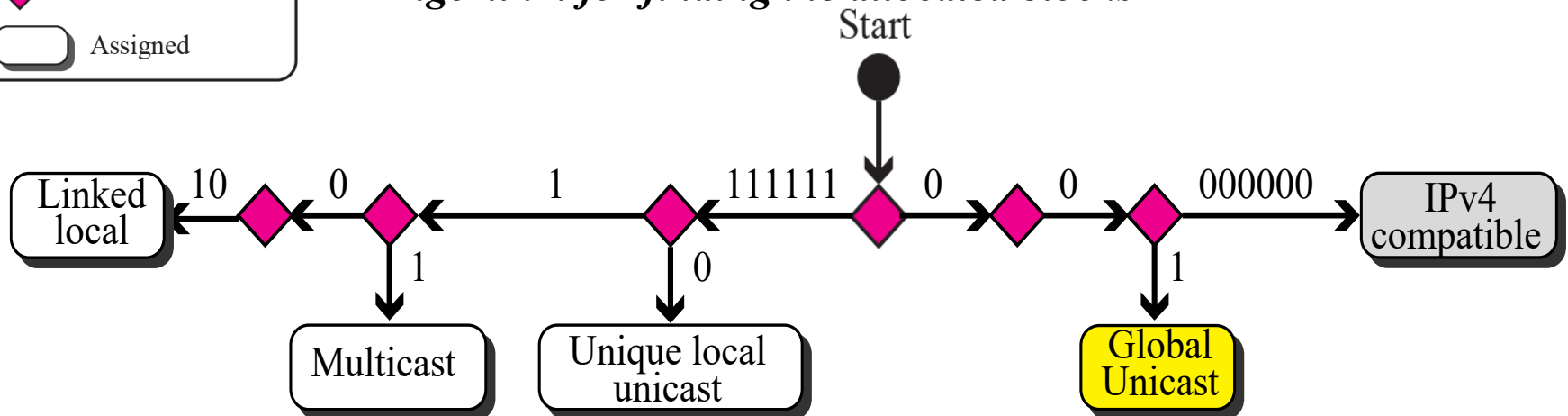
	Block Prefix	CIDR	Block Assignment	Fraction
1	0000 0000	0000::/8	Reserved (IPv4 compatible)	1/256
	0000 0001	0100::/8	Reserved	1/256
	0000 001	0200::/7	Reserved	1/128
	0000 01	0400::/6	Reserved	1/64
	0000 1	0800::/5	Reserved	1/32
	0001	1000::/4	Reserved	1/16
2	001	2000::/3	Global unicast	1/8
3	010	4000::/3	Reserved	1/8
4	011	6000::/3	Reserved	1/8
5	100	8000::/3	Reserved	1/8
6	101	A000::/3	Reserved	1/8
7	110	C000::/3	Reserved	1/8
8	1110	E000::/4	Reserved	1/16
	1111 0	F000::/5	Reserved	1/32
	1111 10	F800::/6	Reserved	1/64
	1111 110	FC00::/7	Unique local unicast	1/128
	1111 1110 0	FE00::/9	Reserved	1/512
	1111 1110 10	FE80::/10	Link local addresses	1/1024
	1111 1110 11	FEC0::/10	Reserved	1/1024
	1111 1111	FF00::/8	Multicast addresses	1/256

Legend

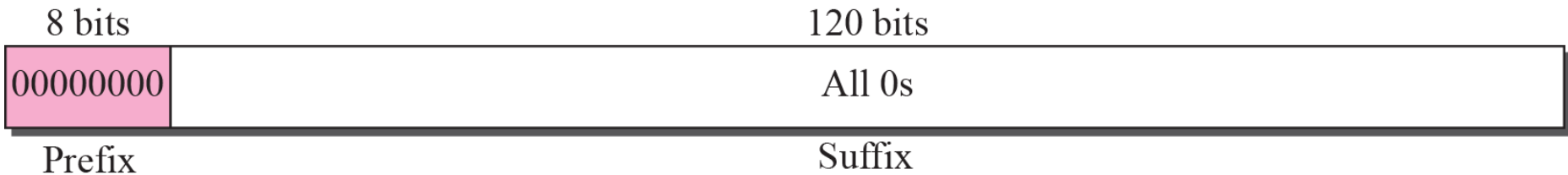
◆ Check next bit or bits

▭ Assigned

Algorithm for finding the allocated blocks



Unspecified address



***The unspecified address in IPv6 is ::/128.
It should never be used as a destination
address.***

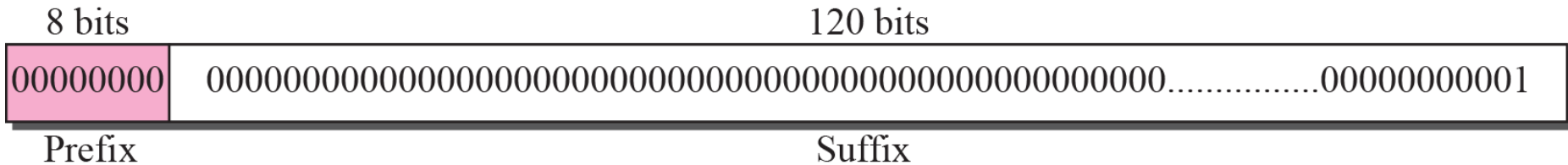
Example

Comparing the unspecified address in IPv4 to the unspecified addresses in IPv6.

Solution

In both architectures, an unspecified address is an all-zero address.

**In IPv4 this address is part of class A address;
in IPv6 this address is part of the reserved block.**



***The loopback address in IPv6 is ::1/128.
It should never be used as a
destination address.***

Compare the loop addresses in IPv4 to the loopback address in IPv6.

Solution

There are two differences in this case.

In IPv4, in classful addressing,

- **a whole block is allocated for loopback addresses;**
- **loopback block is part of the class A block.**

□ in IPv6

- **only one address is allocated as the loopback address.**
- **it is only one single address in the reserved block.**

Embedded IPv4 Address

During the transition from IPv4 to IPv6, hosts can use their IPv4 addresses embedded in IPv6 addresses.

Two formats have been designed for this purpose: **compatible** and **mapped**.

A **compatible address** is an address of **96 bits of zero** followed by **32 bits of IPv4** address.

It is used when a computer using IPv6 wants to send a message to another computer using IPv6. But packet passes through a region where the networks are still using IPv4.

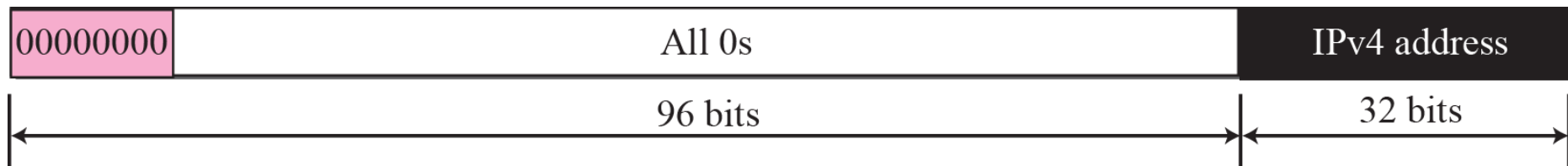
For example,

⇒ the IPv6 address **0::2.13.17.14** becomes **2.13.17.14** in IPv4

⇒ Alternatively, The IPv4 address is prepended with 96 zeros to create a 128-bit IPv6 address.

⇒ This subblock is a reservation that can contain up to 2^{32} addresses.

⇒ The CIDR notation for this subblock is **::/96**.



Compatible Address

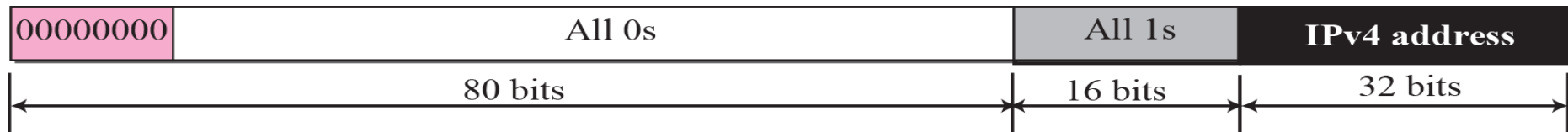
Mapped address

A **mapped address** is used when a computer that has migrated to IPv6 wants to send a packet to a computer still using IPv4. The packet travels mostly through IPv6 networks but is finally delivered to a host that uses IPv4.

It comprises of 80 bits of zero, followed by 16 bits of one, followed by the 32-bit IPv4 address.

For example, the IPv4 address **2.13.17.14** becomes **0::FFFF:2.13.17.14**

The IPv4 address is prepended with 16 ones and 80 zeros to create a 128-bit IPv6 address.



mapped and compatible addresses are designed such that when calculating the checksum, one can use either the embedded address or the total address because extra 0s or 1s in multiples of 16 do not have any effect in checksum calculation.

This is important for UDP and TCP, which use a pseudoheader to calculate the checksum because the checksum calculation is not affected if the address of the packet is changed from IPv6 to IPv4 by a router.

Ex 1000
1111 => 1000

0010
1111 => 0010

IPv4 and IPv6 coexistence

Dual Stack

In dual stack networking, all network devices including router, servers, and firewall will be configured for both IPv4 and IPv6 capabilities.

The whole network must understand both IP versions packets and must allow processing of IPv4 and IPv6 data traffic at the same time. Dual stack devices run both IPv4 and IPv6 protocol stacks simultaneously.

Tunneling

In tunneling IPv6 packet is transporting over an IPv4 network. The dual stack routers encapsulate the IPv6 packet inside into the IPv4 packet, just like the **router** encapsulates other data into IPv4 packets.

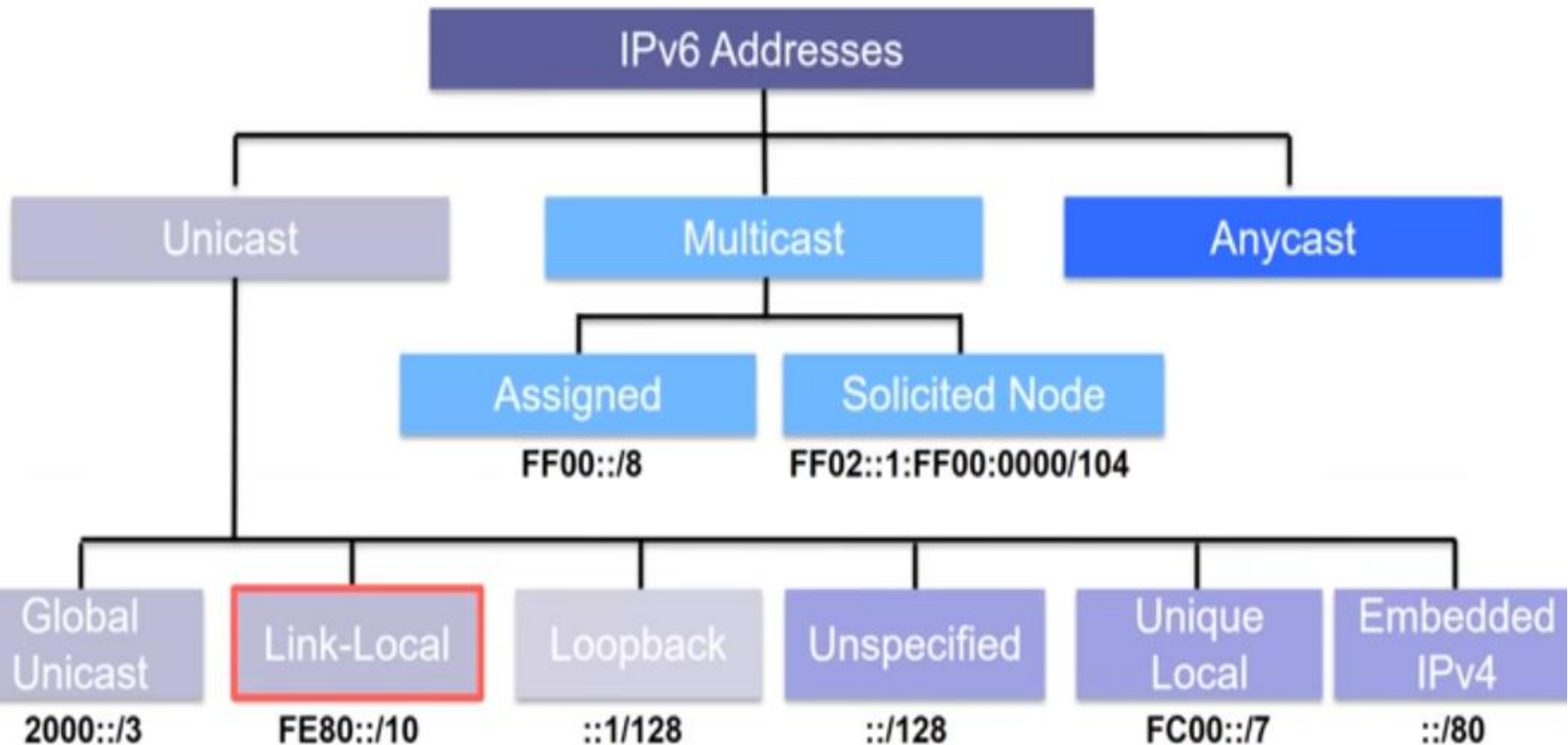
We can communicate using tunneling method, with isolated IPv6 networks without upgrading the IPv4 infrastructure. We only required tunnels configuration between border routers or between a border router and a host however, the important point is that both tunnel endpoints must support both IPv4 and IPv6 protocol stacks.

Translation

Network Address Translation 64 (NAT64) allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4.

An IPv6 packet is translated to an IPv4 packet and vice versa

IPv6 Address Types



IPv6 does not have a “broadcast” address.

Unique local unicast address

Unique Local Unicast Block (prefix - **FC00::/7**)

Some blocks in the IPv4 address space were reserved for **private addressing**.

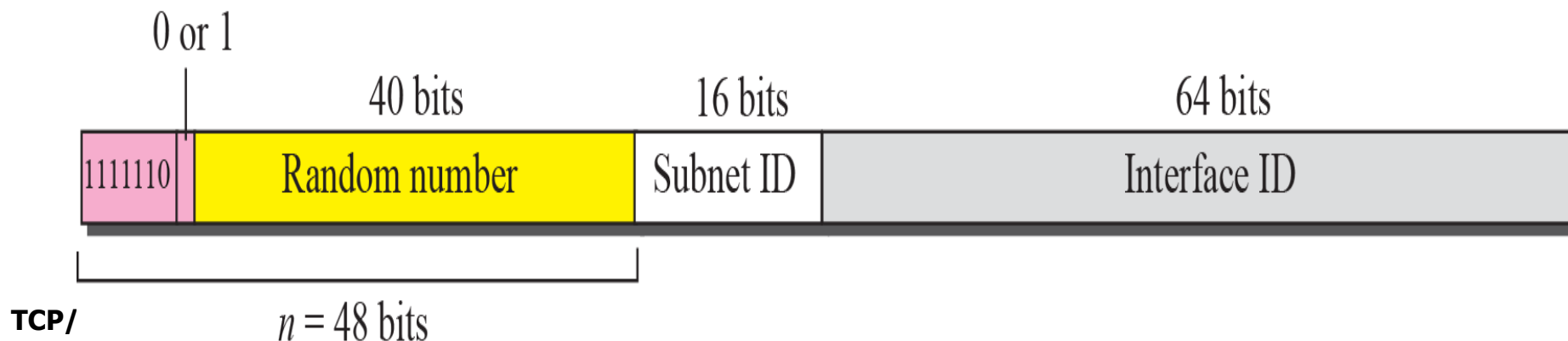
IPv6 uses two large blocks for private addressing:

The packet carrying this type of address as the destination address is not expected to be routed.

This type of address has the block identifier **1111 110**, the next bit can be 0 or 1 to define how the address is selected (locally or by an authority).

The **next 40 bits** are selected by the site using a randomly generated number of length 40 bits.

This means that the total of 48 bits defines a subblock that looks like a global unicast



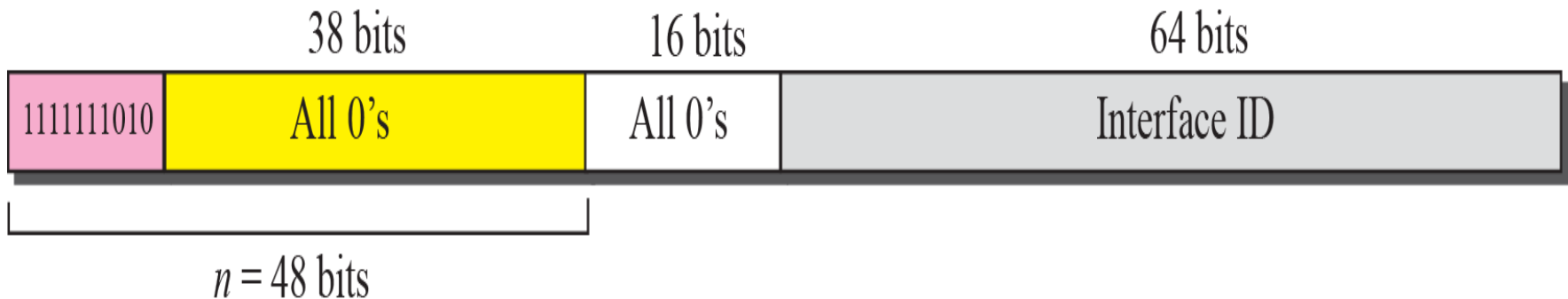
link local block. (prefix - **FE80::/10**)

A subblock in this block can be used as a private address in a network.

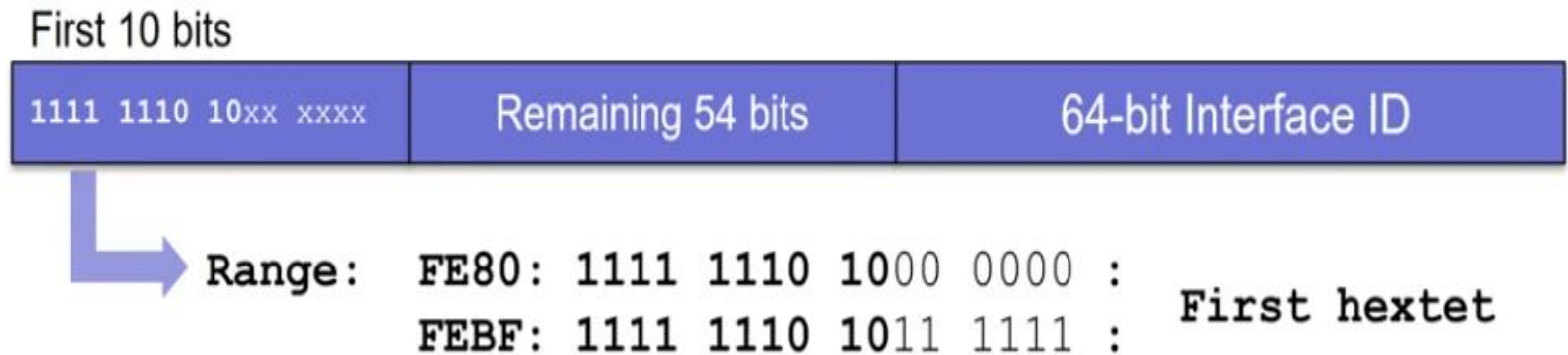
This type of address has the block identifier **111111010**.

The **next 54** bits are set to zero.

The last 64 bits can be changed to define the interface for each computer



Link-Local Unicast Range



- **Link** – Network segment
- Link-local means, local to that link or network.

Analogous to APIPA

APIPA assigns a class B IP address from 169.254. 0.0 to 169.254. 255.255 to the client when a DHCP server is either permanently or temporarily unavailable.

Designed for small non-routable networks, if a DHCP server becomes available later, the APIPA address is replaced with one from the DHCP server.

```
PC> ipconfig
```

```
Windows IP Configuration
```

```
Ethernet adapter Local Area Connection:
```

```
    Connection-specific DNS Suffix  :
```

```
    Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
```

```
    IPv4 Address. . . . . : 192.168.1.101
```

```
    Subnet Mask . . . . . : 255.255.255.0
```

```
    Default Gateway . . . . . : 192.168.1.1
```

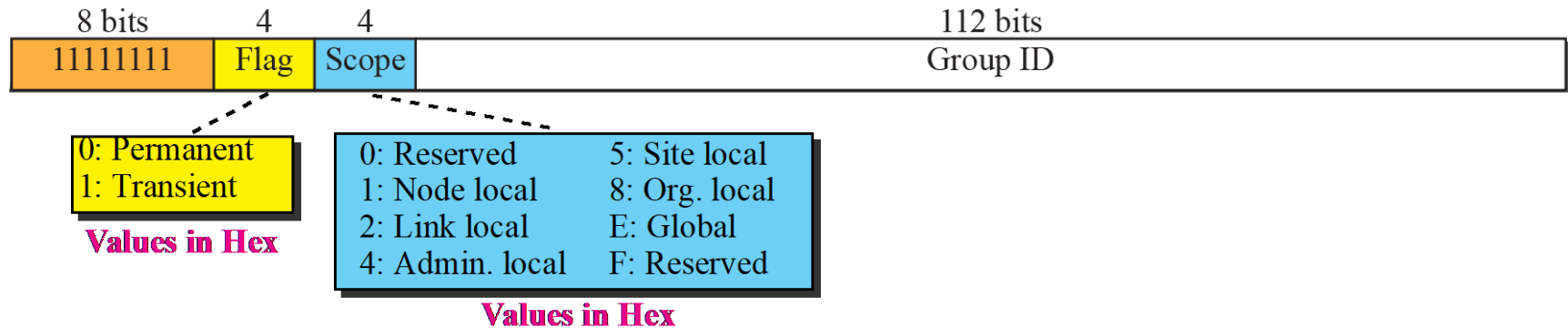
- IPv6 is automatically enabled with Windows, MAC and Linux operating systems.

Multicast address

Multicast addresses are used to define a group of hosts instead of just one.

In IPv6 a large block of addresses are assigned for multicasting. All these addresses use the prefix **11111111**. (prefix - **FF00::/8**)

The second field is a flag that defines the group address as either permanent or transient.



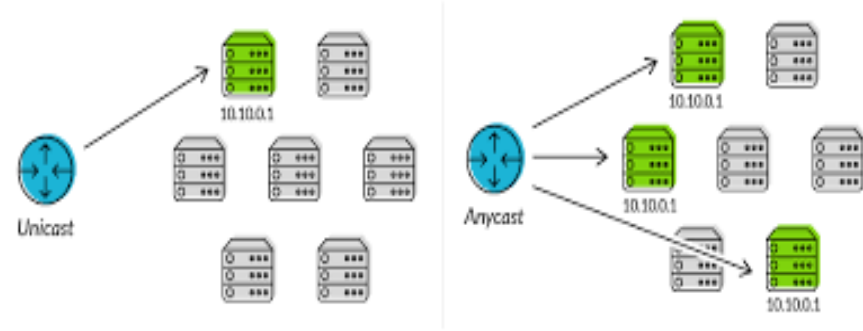
A permanent group address is defined by the Internet authorities and can be accessed at all times.

A transient group address is used only temporarily.

for example, Systems engaged in a teleconference, can use a transient group address.

Anycast

Anycast is a network addressing and routing method in which incoming requests can be routed to a variety of different locations or “nodes.”



In the context of a CDN, Anycast typically routes incoming traffic to the nearest data center with the capacity to process the request efficiently.

Anycast Vs Multicast

In Anycast, the data packet is received by the nearest host, while Multicast ensures simultaneous delivery to multiple interested recipients.

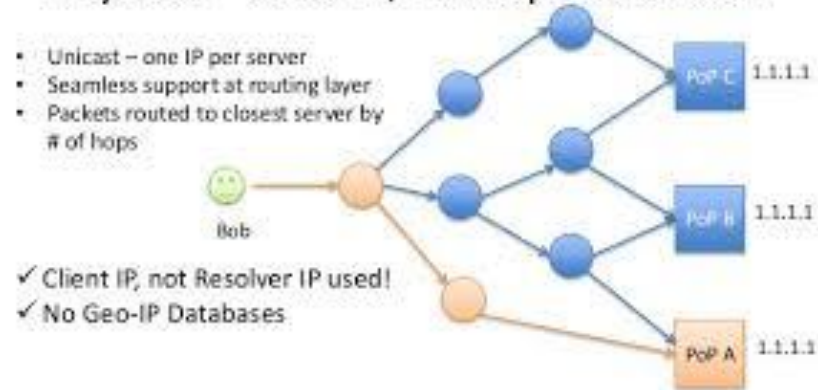
Anycast Vs Unicast

An IPv6 unicast address is supported on a single interface on one node,

An IPv6 anycast address is assigned to a group of interfaces on multiple nodes.

Anycast – One IP, Multiple Servers

- Unicast – one IP per server
- Seamless support at routing layer
- Packets routed to closest server by # of hops



GLOBAL UNICAST ADDRESSES

This block in the address space that is used for unicast (one-to-one) communication between two hosts in the Internet is called global unicast address block.

CIDR notation for the block is **2000::/3**, which means that the three leftmost bits are the same for all addresses in this block (**001**).

The size of this block is **2^{125} bits**, which is more than enough for the Internet expansion in the many years to come.

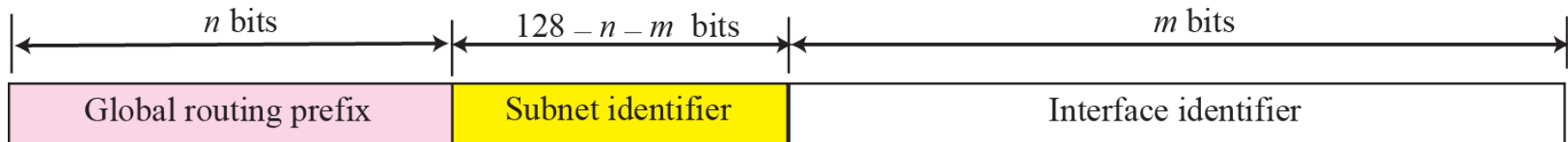


Table 26.2 *Recommended Length of Different Parts in Unicast Addressing*

<i>Block Assignment</i>	<i>Length</i>
Global routing prefix (n)	48 bits
Subnet identifier ($128 - n - m$)	16 bits
Interface identifier (m)	64 bits

Parts of a Global Unicast Address

IPv4 Unicast Address

/?



IPv6 Global Unicast Address

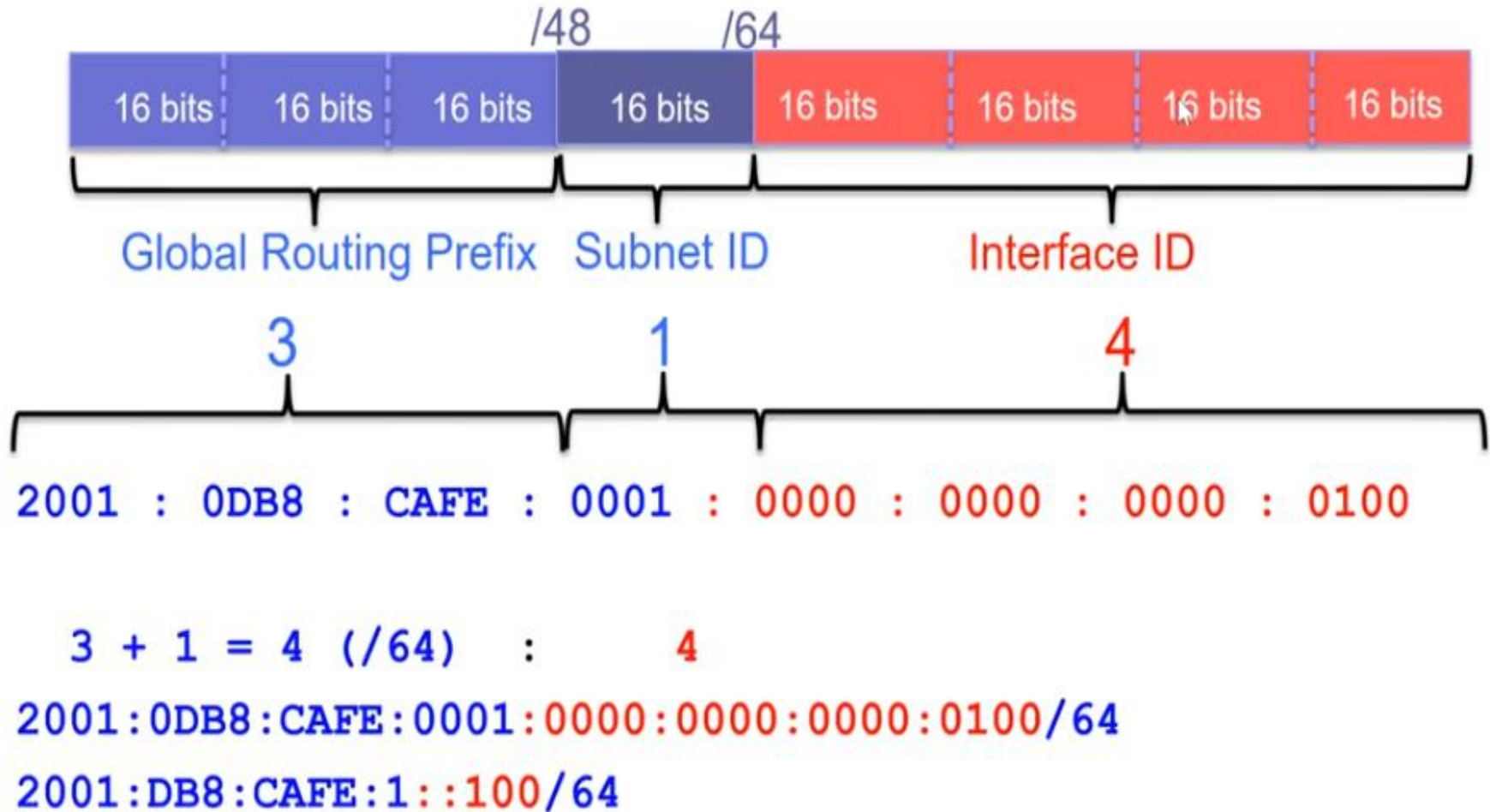
/48

/64



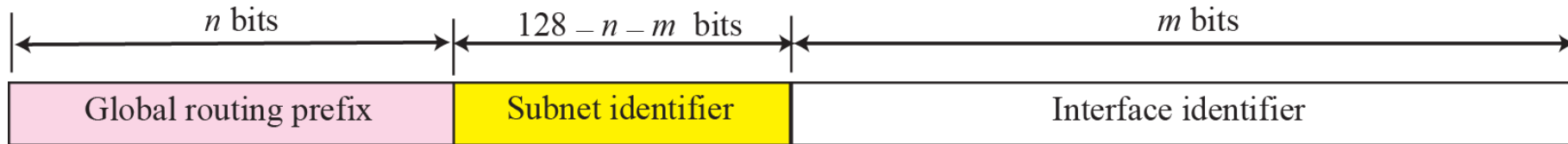
- 64-bit Interface ID = 18 quintillion (18,446,744,073,709,551,616) devices/subnet
- 16-bit Subnet ID (initially recommended) = 65,536 subnets

/64 Global Unicast Address and the 3-1-4 Rule



Interface Identifier

The last 64 bits define the interface identifier. The interface identifier is similar to hostid



In IPv4 addressing, there is not a specific relation between the hostid (at the IP level) and physical or MAC address (at the data link layer) because the physical address is normally much longer than the hostid.

For example, using the Ethernet technology, the physical address is 48 bits while the hostid is less than 32 bits.

The IPv6 addressing,

A physical address whose length is less than 64 bits can be embedded as the whole or part of the interface identifier, eliminating the mapping process.

Two common physical addressing schemes used are :

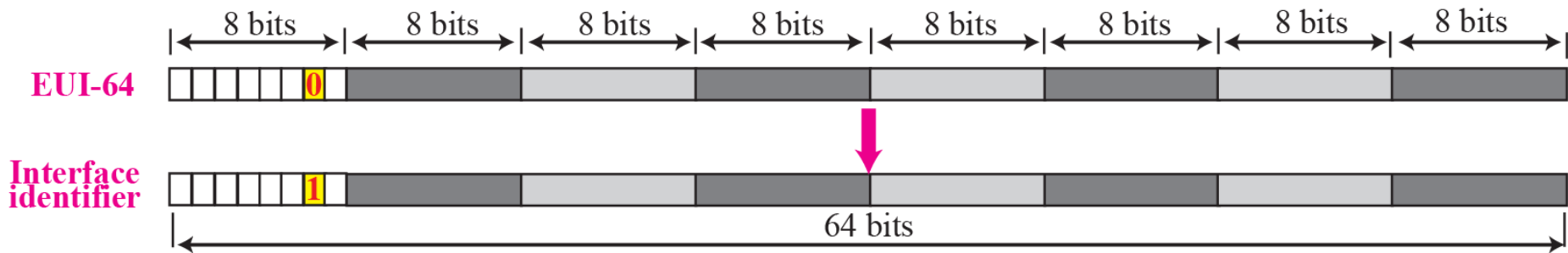
- The 64-bit extended unique identifier (EUI-64) defined by IEEE
- 48-bit physical address defined by Ethernet.

Interface Identifier

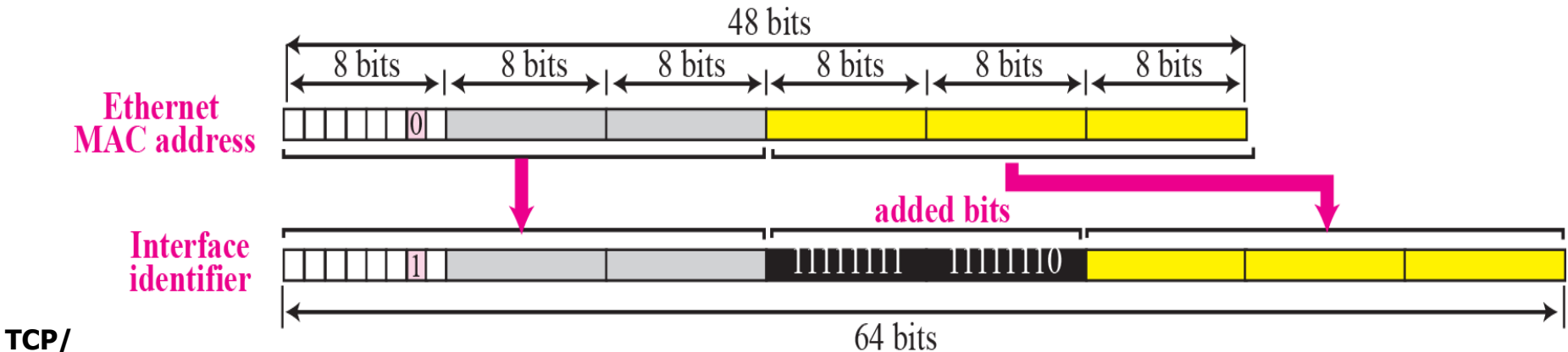
EUI-64 - 64-bit extended unique identifier defined by IEEE

Ethernet MAC - 48-bit physical address defined by Ethernet

Mapping for EUI-64



Mapping for Ethernet MAC



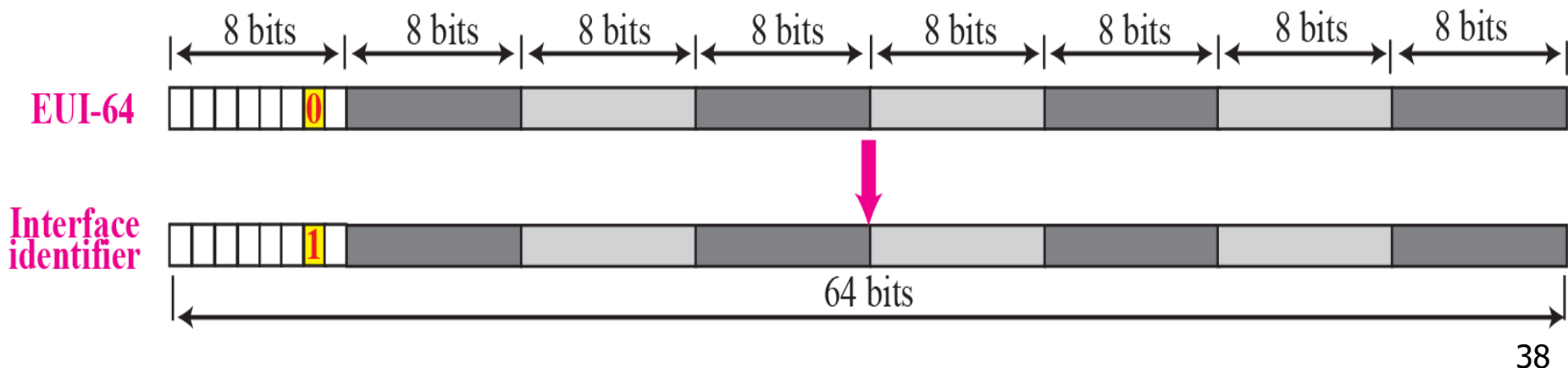
Example

Find the interface identifier if the physical address in the EUI is
(F5-A9-23-EF-07-14-7A-D2)₁₆
using the format we defined for Ethernet addresses.

Solution

We only need to change the seventh bit of the first octet from 0 to 1 and change the format to colon hex notation.

The result is **F7A9:23EF:0714:7AD2**



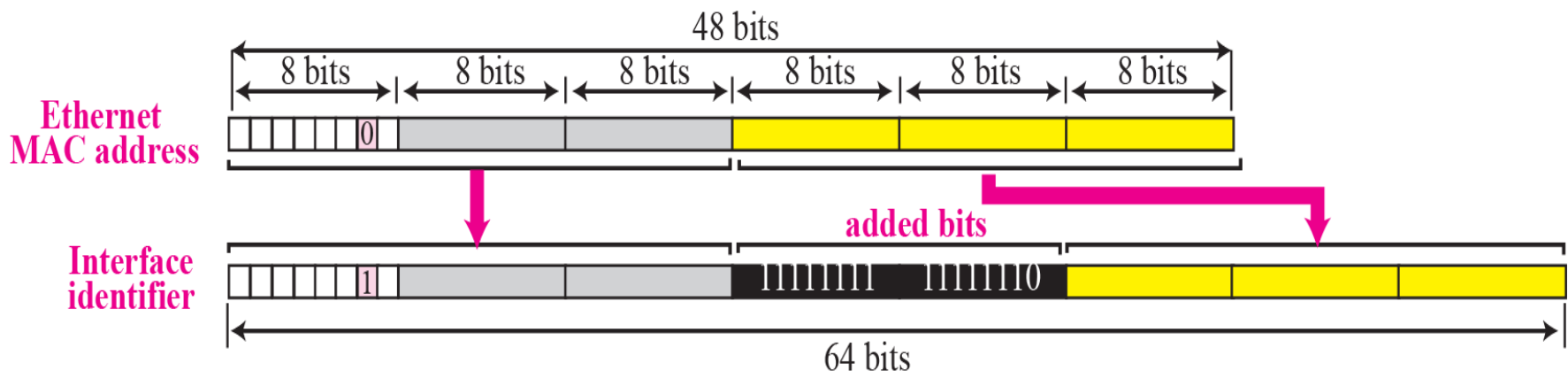
Example

Find the interface identifier if the Ethernet physical address is
(F5-A9-23-14-7A-D2)₁₆
using the format we defined for Ethernet addresses.

Solution

- ✓ We only need to change the seventh bit of the first octet from 0 to 1,
- ✓ insert two octet FFFE16 and change the format to colon hex notation.

The result is **F7A9:23FF:FE14:7AD2.**



Example

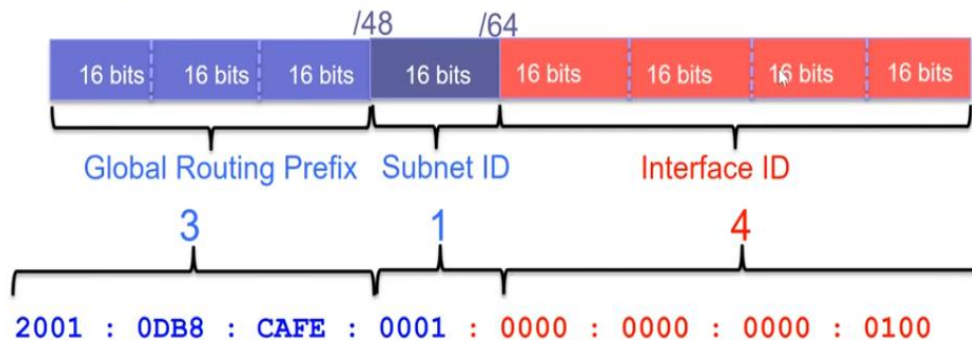
An organization is assigned the block
2000:1456:2474/48.

What is the CIDR notation for the blocks in the first and second subnets in this organization.

Solution

Theoretically, the first and second subnets should use the block with subnet identifier 0000_{16} and 0001_{16} .

/64 Global Unicast Address and the 3-1-4 Rule



This means that the blocks are

2000:1456:2474:0000/64

and

2000:1456:2474:0001/64.

$3 + 1 = 4$ (/64) : 4

2001:0DB8:CAFE:0001:0000:0000:0000:0100/64

2001:DB8:CAFE:1::100/64

Example

An organization is assigned the block
2000:1456:2474/48

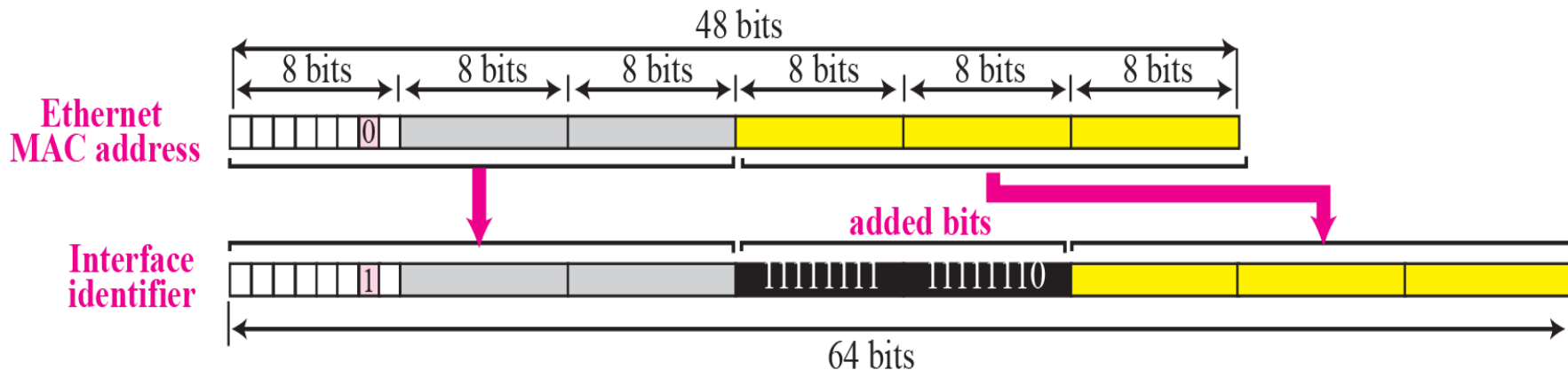
What is the IPv6 address of an interface in the **fourth subnet**
if the IEEE physical address is **(F5-A9-23-14-7A-D2)₁₆**.

Solution

The interface identifier is **F7A9:23FF:FE14:7AD2**.

If we add this identifier to the global prefix and the subnet identifier,
we get:

2000:1456:2474:0003:F7A9:23FF:FE14:7AD2/128



In IPv6 host can configure itself

DHCP protocol can still be used to allocate an IPv6 address to a host

Example

Assume a host with Ethernet address
(F5-A9-23-11-9B-E2)₁₆ has joined the network.

What would be its global unicast address if the

- ❑ global unicast prefix of the organization is **3A21:1216:2165** and
- ❑ subnet identifier is **A245:1232**.

Solution : The host first creates its interface identifier as

F5-A9-23-11-9B-E2 → F7A9:23FF:FE11:9BE2

using the Ethernet address read from its card.

The host then creates its link-local address as

FE80::F7A9:23FF:FE11:9BE2

Example *Continued*

Assuming that this address is unique, the host sends a router solicitation message and receives the router advertisement message that announces the combination of global unicast prefix and the subnet identifier as

3A21:1216:2165:A245:1232.

The host then appends its interface identifier to this prefix to find and store its global unicast address as:

3A21:1216:2165:A245:1232:F7A9:23FF:FE11:9BE2